



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

Course Title:	Fundamentals of Data Engineering
Course Number:	COE848
Semester/Year (e.g. F2016)	W2026

Instructor:	Dr. Yasaman Ahmadiadli
Lab Report Number	Lab 5
Lab Title	Lab 5: Database Programming

Due Date	March 26, 2026
Submission Date	March 26, 2026

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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: <http://www.ryerson.ca/senate/current/pol60.pdf>

How to Compiling/Running/Building:

Before getting started, this lab does require certain support files and Dependencies:

- Java JDK: For the Java program to work, a more recent JDK version must be installed
- JDBC driver: The application requires connection and communication between the PCs SQLite and Java programs; as such a certain JDBC Driver(sqlite-jdbc-3.51.3.0.jar) is required. Note that this folder must be in the exact same directory as the database file and java package folder, to be compatible with the others
- Do note that SQLite must be active as well for this to proceed

File Structure:

The working folder must include the database file, the JDBC driver(.jar) file, and the Java file that is stored within its package folder(in this case, RestaurantDatabaseJava.java is stored within the package restaurantdatabasejava, as such its directory is restaurantdatabasejava/RestaurantDatabaseJava.java).

How to run it:

Now that we have the right folder structure, in order to run the program, we could simply use an IDE like netbeans, or run it directly using a CMD or any Command Line Interface:

- Running with IDE: To run with IDE, we must first import the java package folder into the IDE. for Netbeans the original entire project folder is kept and can be used. Either way, the IDE and java must be configured properly, such that the java file exists within the package and is set properly. Now the sqlite-jdbc-3.51.3.0.jar file must be imported, by placing it within the project folder's library manually and can be imported directly from the IDE by configuring the projects library inside the IDE; making sure you do it both ways ensures that this step is done correctly.
- Running with CMD: You can even run with the Command Line Interface. By simply taking the current project folder structure as outlined above, we simply open a terminal/CMD from that certain folder with all the files(JDBC driver, Java package folder, database file). Then we can type:

```
javac -cp ".;sqlite-jdbc-3.51.3.0.jar" restaurantdatabasejava/RestaurantDatabaseJava.java
java -cp ".;sqlite-jdbc-3.51.3.0.jar" restaurantdatabasejava.RestaurantDatabaseJava
```

By doing so, we compile the java file and then we run it, again with respect to its package folder.

```

PS C:\Users\hasib\Documents\Coe848 - Labs_Databases\HasibBhu
iyan_Lab5> javac -cp ".;sqlite-jdbc-3.51.3.0.jar" restaurant
databasejava/RestaurantDatabaseJava.java
PS C:\Users\hasib\Documents\Coe848 - Labs_Databases\HasibBhu
iyan_Lab5> java -cp ".;sqlite-jdbc-3.51.3.0.jar" restaurantd
atabasejava.RestaurantDatabaseJava

--- Restaurant Database Menu ---
1. Restaurants with total capacity > 80
2. Restaurants with Chicken based menu items
3. Tables available for a group of 4 people
4. Highest priced menu item in each restaurant respectively
5. Count Japanese restaurants
6. Highest paid employee in a Mexican Restaurant and a India
n Restaurant
7. Operating hours of Villa Grill
8. Restaurant with highest capacity
9. Items in Villa Grill that are less than $10
10. Check for table accomodating for 6-8 people in Villa Gri
ll
11. Add Customer
12. Delete OrderHandledBy entry
13. Update Customer phone
14. Exit

```

Issues and Cautions that could potentially arise:

- Database Path: The java file links to the database by literally using the file path for the database(the absolute datapath), **please change this filepath to match the one being used:**
 "jdbc:sqlite:C:/Users/hasib/Documents/Coe848 - Labs_Databases/restaurant.db"
- Input Errors: Any input errors or inputs of wrong data type will be terminated and will display an InputMismatchException error, the program must be restarted.

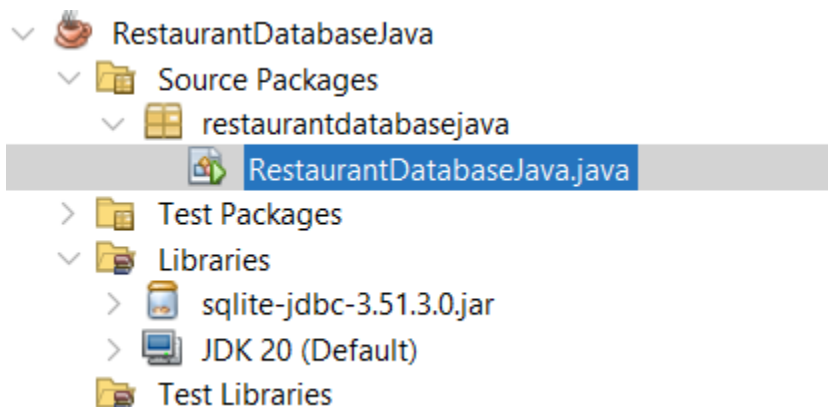


Figure 1: IDE setup with sqlite-jdbc-3.51.3.0.jar

```

--- Restaurant Database Menu ---
1. Restaurants with total capacity > 80
2. Restaurants with Chicken based menu items
3. Tables available for a group of 4 people
4. Highest priced menu item in each restaurant respectively
5. Count Japanese restaurants
6. Highest paid employee in a Mexican Restaurant and a Indian Restaurant
7. Operating hours of Villa Grill
8. Restaurant with highest capacity
9. Items in Villa Grill that are less than $10
10. Check for table accomodating for 6-8 people in Burger Barn
11. Add Customer
12. Delete OrderHandledBy entry
13. Update Customer phone
14. Exit
|

```

Figure 2: Restaurant Database Menu

1

```

--- RESULTS ---
name : Burger Barn | total_capacity : 90 |
-----
name : Seaside Grill | total_capacity : 85 |
-----

```

Figure 3: User input 1: Any restaurant with total capacity over 80

2

```

--- RESULTS ---
name : Spice Route | item_name : Butter Chicken |
-----
name : Taco Town | item_name : Chicken Taco |
-----
name : Dragon Wok | item_name : Sweet and Sour Chicken |
-----

```

Figure 4: User input 2: Any restaurant with “Chicken” based menu items

3

```
--- RESULTS ---
restaurant_name : Villa Grill | table_number : 1 | seating_capacity : 4 |
-----
restaurant_name : Villa Grill | table_number : 7 | seating_capacity : 6 |
-----
restaurant_name : Sunset Sushi | table_number : 3 | seating_capacity : 4 |
-----
restaurant_name : Pasta Palace | table_number : 5 | seating_capacity : 4 |
-----
restaurant_name : Spice Route | table_number : 2 | seating_capacity : 6 |
-----
restaurant_name : Taco Town | table_number : 4 | seating_capacity : 4 |
-----
restaurant_name : Burger Barn | table_number : 8 | seating_capacity : 6 |
-----
restaurant_name : Seaside Grill | table_number : 9 | seating_capacity : 4 |
-----
```

Figure 5: User input 3: Tables available and suitable for a group of 4 people

4

```
--- RESULTS ---
item_name : Mango Juice | price : 5.0 | restaurant_name : Villa Grill |
-----
item_name : Salmon Sushi Roll | price : 14.0 | restaurant_name : Sunset Sushi |
-----
item_name : Spaghetti Carbonara | price : 16.0 | restaurant_name : Pasta Palace |
-----
item_name : Butter Chicken | price : 17.0 | restaurant_name : Spice Route |
-----
item_name : Chicken Taco | price : 9.0 | restaurant_name : Taco Town |
-----
item_name : Sweet and Sour Chicken | price : 13.0 | restaurant_name : Dragon Wok |
-----
item_name : Double Cheeseburger | price : 12.0 | restaurant_name : Burger Barn |
-----
item_name : Vegan Salad Bowl | price : 11.0 | restaurant_name : Green Garden |
-----
item_name : Grilled Lobster | price : 24.0 | restaurant_name : Seaside Grill |
-----
item_name : Shawarma Platter | price : 15.0 | restaurant_name : Villa Grill |
-----
```

Figure 6: User input 4: Highest priced menu item in each available store

5

```
--- RESULTS ---
```

```
JapaneseRestaurant_Count : 1 |
```

```
-----
```

Figure 7: User input 5: Any “Japanese” based restaurant as dictated by cuisine type gets counted

6

```
--- RESULTS ---
```

```
first_name : Raj | last_name : Patel | hourly_wage : 29.0 | restaurant_name : Spice Route |
```

```
-----
```

```
first_name : Carlos | last_name : Lopez | hourly_wage : 18.0 | restaurant_name : Taco Town |
```

```
-----
```

Figure 8: User input 6: Highest paid employee in a Mexican based restaurant and the highest paid in an Indian based restaurant

7

```
--- RESULTS ---
```

```
name : Villa Grill | opening_hours : 10:00 | Closing_hours : 22:00 |
```

```
-----
```

Figure 9: User input 7: Operating Hours of Villa Grill restaurant

8

```
--- RESULTS ---
```

```
name : Burger Barn | total_capacity : 90 |
```

```
-----
```

Figure 10: User input 8: The restaurant with the highest capacity

9

```
--- RESULTS ---
```

```
restaurant_name : Villa Grill | item_name : Mango Juice | price : 5.0 |
```

```
-----
```

Figure 11: User input 9: The items in Villa Grill that are below \$10

10

```
--- RESULTS ---  
name : Villa Grill | table_number : 7 |  
-----
```

Figure 12: User input 10: Check for a table to accommodate for 6-8 people in Villa Grill

11

```
--- ADDING CUTOMER ---  
First name:  
TEST  
Last name:  
TEST  
Email:  
TEST@email.com  
Customer added.
```

Figure 13: User input 11: Add a “customer” to the Customer Table

12

```
--- DELETING ORDER ---  
Enter Order ID:  
123  
Enter Staff ID:  
1234  
Deleted.
```

Figure 14: User input 12: Delete a specific order in OrderHandledBy Table(Orders in progress and being made are recorded in a separate table, the OrderHandledBy table, that makes entries of active and processing orders that have active order numbers and current staff members in preparation of it)

13

```
--- UPDATING CUSTOMER PHONE ---  
Enter Customer ID:  
234  
New phone number:  
4544445445  
Updated.
```

Figure 15: User input 13: Updating the “customer” phone number within the Customer table

14

```
Goodbye!
```

Figure 16: User input 14: EXIT or Terminate program

```
PRAGMA foreign_keys=OFF;  
BEGIN TRANSACTION;  
CREATE TABLE Restaurant (  
    restaurant_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    name TEXT NOT NULL,  
    address TEXT NOT NULL,  
    phone_number TEXT NOT NULL,  
    cuisine_type TEXT,  
    opening_hours TEXT,  
    Closing_hours TEXT,  
    total_capacity INTEGER  
);  
INSERT INTO Restaurant VALUES(1,'Villa Grill','123 King St  
Toronto','4165551000','Mediterranean','10:00','22:00',80);  
INSERT INTO Restaurant VALUES(2,'Sunset Sushi','22 Queen St  
Toronto','4165551001','Japanese','11:00','23:00',60);  
INSERT INTO Restaurant VALUES(3,'Pasta Palace','77 Dundas St
```



```

Toronto','4165551002','Italian','10:00','22:00',70);
INSERT INTO Restaurant VALUES(4,'Spice Route','90 Bloor St
Toronto','4165551003','Indian','11:00','23:00',75);
INSERT INTO Restaurant VALUES(5,'Taco Town','15 Front St
Toronto','4165551004','Mexican','10:00','21:00',50);
INSERT INTO Restaurant VALUES(6,'Dragon Wok','55 Yonge St
Toronto','4165551005','Chinese','11:00','22:30',65);
INSERT INTO Restaurant VALUES(7,'Burger Barn','300 College St
Toronto','4165551006','American','10:00','22:00',90);
INSERT INTO Restaurant VALUES(8,'Green Garden','88 Bay St
Toronto','4165551007','Vegetarian','09:00','21:00',55);
INSERT INTO Restaurant VALUES(9,'Seaside Grill','12 Harbour St
Toronto','4165551008','Seafood','11:00','23:00',85);
INSERT INTO Restaurant VALUES(10,'Golden Curry','9 Spadina Ave
Toronto','4165551009','Thai','11:00','22:00',60);
CREATE TABLE Customer (
    customer_id INTEGER PRIMARY KEY AUTOINCREMENT,
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    email TEXT UNIQUE,
    phone_number TEXT,
    loyalty_status TEXT
);
INSERT INTO Customer VALUES(1,'Bob','Smith','bob@email.com','4161110001','Gold');
INSERT INTO Customer VALUES(2,'Alice','Brown','alice@email.com','4161110002','Silver');
INSERT INTO Customer VALUES(3,'John','Lee','john@email.com','4161110003','None');
INSERT INTO Customer VALUES(4,'Maria','Garcia','maria@email.com','4161110004','Gold');
INSERT INTO Customer VALUES(5,'David','Kim','david@email.com','4161110005','Silver');
INSERT INTO Customer VALUES(6,'Sara','Wilson','sara@email.com','4161110006','None');
INSERT INTO Customer VALUES(7,'Omar','Hassan','omar@email.com','4161110007','Gold');
INSERT INTO Customer VALUES(8,'Lina','Patel','lina@email.com','4161110008','Silver');
INSERT INTO Customer VALUES(9,'Chris','Johnson','chris@email.com','4161110009','None');
INSERT INTO Customer
VALUES(10,'Emma','Taylor','emma@email.com','4161110010','Gold');
INSERT INTO Customer
VALUES(12,'JAVA_INSERTION_TEST','JAAV_INSERTION_TEST','java@email.com','4166
144160',NULL);
INSERT INTO Customer VALUES(13,'TEST','TEST','TEST@email.com',NULL,NULL);
CREATE TABLE DiningTable (
    restaurant_id INTEGER NOT NULL,
    table_number INTEGER NOT NULL,
    seating_capacity INTEGER NOT NULL,
    availability_status TEXT,
    seating_location_description TEXT,
    PRIMARY KEY (restaurant_id, table_number),
    FOREIGN KEY (restaurant_id)

```

```

REFERENCES Restaurant(restaurant_id)
ON DELETE CASCADE
);
INSERT INTO DiningTable VALUES(1,1,4,'Available','Window');
INSERT INTO DiningTable VALUES(1,7,6,'Available','Patio');
INSERT INTO DiningTable VALUES(2,3,4,'Available','Center');
INSERT INTO DiningTable VALUES(3,5,4,'Available','Corner');
INSERT INTO DiningTable VALUES(4,2,6,'Available','Center');
INSERT INTO DiningTable VALUES(5,4,4,'Available','Patio');
INSERT INTO DiningTable VALUES(6,6,2,'Available','Window');
INSERT INTO DiningTable VALUES(7,8,6,'Available','Booth');
INSERT INTO DiningTable VALUES(8,1,2,'Available','Window');
INSERT INTO DiningTable VALUES(9,9,4,'Available','Patio');
CREATE TABLE Orders (
    order_id INTEGER PRIMARY KEY AUTOINCREMENT,
    customer_id INTEGER NOT NULL,
    restaurant_id INTEGER NOT NULL,
    reservation_id INTEGER,
    order_date TEXT NOT NULL,
    total_amount REAL,
    order_status TEXT,
    FOREIGN KEY (customer_id)
        REFERENCES Customer(customer_id),
    FOREIGN KEY (restaurant_id)
        REFERENCES Restaurant(restaurant_id),
    FOREIGN KEY (reservation_id)
        REFERENCES Reservation(reservation_id)
);
INSERT INTO Orders VALUES(1,1,1,1,'2026-04-09',100.0,'Completed');
INSERT INTO Orders VALUES(2,2,2,2,'2026-04-10',42.0,'Completed');
INSERT INTO Orders VALUES(3,3,3,3,'2026-04-11',32.0,'Completed');
INSERT INTO Orders VALUES(4,4,4,4,'2026-04-11',50.0,'Completed');
INSERT INTO Orders VALUES(5,5,5,5,'2026-04-12',27.0,'Completed');
INSERT INTO Orders VALUES(6,6,6,6,'2026-04-12',26.0,'Completed');
INSERT INTO Orders VALUES(7,7,7,7,'2026-04-13',48.0,'Completed');
INSERT INTO Orders VALUES(8,8,8,8,'2026-04-13',22.0,'Completed');
INSERT INTO Orders VALUES(9,9,9,9,'2026-04-14',72.0,'Completed');
INSERT INTO Orders VALUES(10,10,1,10,'2026-04-15',30.0,'Completed');
CREATE TABLE Staff (
    staff_id INTEGER PRIMARY KEY AUTOINCREMENT,
    restaurant_id INTEGER NOT NULL,
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    role TEXT,
    hourly_wage REAL,
    FOREIGN KEY (restaurant_id)

```

```

REFERENCES Restaurant(restaurant_id)
);
INSERT INTO Staff VALUES(1,1,'Ross','Miller','Manager',30.0);
INSERT INTO Staff VALUES(2,1,'Anna','Scott','Waiter',18.0);
INSERT INTO Staff VALUES(3,2,'Kenji','Tanaka','Chef',28.0);
INSERT INTO Staff VALUES(4,3,'Marco','Rossi','Chef',26.0);
INSERT INTO Staff VALUES(5,4,'Raj','Patel','Manager',29.0);
INSERT INTO Staff VALUES(6,5,'Carlos','Lopez','Waiter',18.0);
INSERT INTO Staff VALUES(7,6,'Li','Wang','Chef',25.0);
INSERT INTO Staff VALUES(8,7,'Mike','Brown','Supervisor',22.0);
INSERT INTO Staff VALUES(9,8,'Nina','Singh','Host',17.0);
INSERT INTO Staff VALUES(10,9,'Peter','Hall','Waiter',18.0);
CREATE TABLE MenuItem (
    menu_item_id INTEGER PRIMARY KEY AUTOINCREMENT,
    staff_id INTEGER NOT NULL,
    item_name TEXT NOT NULL,
    price REAL NOT NULL,
    availability_status TEXT,
    FOREIGN KEY (staff_id)
        REFERENCES Staff(staff_id)
);
INSERT INTO MenuItem VALUES(1,1,'Mango Juice',5.0,'Available');
INSERT INTO MenuItem VALUES(2,3,'Salmon Sushi Roll',14.0,'Available');
INSERT INTO MenuItem VALUES(3,4,'Spaghetti Carbonara',16.0,'Available');
INSERT INTO MenuItem VALUES(4,5,'Butter Chicken',17.0,'Available');
INSERT INTO MenuItem VALUES(5,6,'Chicken Taco',9.0,'Available');
INSERT INTO MenuItem VALUES(6,7,'Sweet and Sour Chicken',13.0,'Available');
INSERT INTO MenuItem VALUES(7,8,'Double Cheeseburger',12.0,'Available');
INSERT INTO MenuItem VALUES(8,9,'Vegan Salad Bowl',11.0,'Available');
INSERT INTO MenuItem VALUES(9,10,'Grilled Lobster',24.0,'Available');
INSERT INTO MenuItem VALUES(10,2,'Shawarma Platter',15.0,'Available');
CREATE TABLE OrderHandledBy (
    order_id INTEGER NOT NULL,
    staff_id INTEGER NOT NULL,
    PRIMARY KEY (order_id, staff_id),
    FOREIGN KEY (order_id)
        REFERENCES Orders(order_id),
    FOREIGN KEY (staff_id)
        REFERENCES Staff(staff_id)
);
INSERT INTO OrderHandledBy VALUES(1,2);
INSERT INTO OrderHandledBy VALUES(2,3);
INSERT INTO OrderHandledBy VALUES(3,4);
INSERT INTO OrderHandledBy VALUES(4,5);
INSERT INTO OrderHandledBy VALUES(5,6);
INSERT INTO OrderHandledBy VALUES(6,7);

```

```

INSERT INTO OrderHandledBy VALUES(7,8);
INSERT INTO OrderHandledBy VALUES(8,9);
INSERT INTO OrderHandledBy VALUES(9,10);
INSERT INTO OrderHandledBy VALUES(10,2);
CREATE TABLE OrderContainsItem (
    order_id INTEGER NOT NULL,
    menu_item_id INTEGER NOT NULL,
    quantity INTEGER NOT NULL,
    PRIMARY KEY (order_id, menu_item_id),
    FOREIGN KEY (order_id)
        REFERENCES Orders(order_id),
    FOREIGN KEY (menu_item_id)
        REFERENCES MenuItem(menu_item_id)
);
INSERT INTO OrderContainsItem VALUES(1,1,5);
INSERT INTO OrderContainsItem VALUES(2,2,3);
INSERT INTO OrderContainsItem VALUES(3,3,2);
INSERT INTO OrderContainsItem VALUES(4,4,3);
INSERT INTO OrderContainsItem VALUES(5,5,2);
INSERT INTO OrderContainsItem VALUES(6,6,2);
INSERT INTO OrderContainsItem VALUES(7,7,4);
INSERT INTO OrderContainsItem VALUES(8,8,2);
INSERT INTO OrderContainsItem VALUES(9,9,3);
INSERT INTO OrderContainsItem VALUES(10,10,2);
CREATE TABLE Reservation (
    reservation_id INTEGER PRIMARY KEY AUTOINCREMENT,
    customer_id INTEGER NOT NULL,
    restaurant_id INTEGER NOT NULL,
    table_number INTEGER,
    reservation_date TEXT NOT NULL,
    reservation_time TEXT NOT NULL,
    number_of_guests INTEGER,
    reservation_status TEXT,
    seating_start_time TEXT,
    seating_end_time TEXT,
    FOREIGN KEY (customer_id)
        REFERENCES Customer(customer_id),
    FOREIGN KEY (restaurant_id)
        REFERENCES Restaurant(restaurant_id),
    FOREIGN KEY (restaurant_id, table_number)
        REFERENCES DiningTable(restaurant_id, table_number)
);
INSERT INTO Reservation VALUES(1,1,1,7,'2026-04-09','18:00',5,'Confirmed','18:00','19:30');
INSERT INTO Reservation VALUES(2,2,2,3,'2026-04-10','19:00',3,'Confirmed','19:00','20:30');
INSERT INTO Reservation VALUES(3,3,3,5,'2026-04-11','18:30',2,'Confirmed','18:30','19:30');
INSERT INTO Reservation VALUES(4,4,4,2,'2026-04-11','20:00',4,'Confirmed','20:00','21:30');

```

```
INSERT INTO Reservation VALUES(5,5,5,4,'2026-04-12','17:00',3,'Confirmed','17:00','18:30');
INSERT INTO Reservation VALUES(6,6,6,6,'2026-04-12','19:30',2,'Confirmed','19:30','20:30');
INSERT INTO Reservation VALUES(7,7,7,8,'2026-04-13','18:00',5,'Confirmed','18:00','19:30');
INSERT INTO Reservation VALUES(8,8,8,1,'2026-04-13','17:30',2,'Confirmed','17:30','18:30');
INSERT INTO Reservation VALUES(9,9,9,9,'2026-04-14','20:00',3,'Confirmed','20:00','21:30');
INSERT INTO Reservation
VALUES(10,10,1,1,'2026-04-15','18:30',2,'Confirmed','18:30','19:30');
PRAGMA writable_schema=ON;
CREATE TABLE IF NOT EXISTS sqlite_sequence(name,seq);
DELETE FROM sqlite_sequence;
INSERT INTO sqlite_sequence VALUES('Restaurant',10);
INSERT INTO sqlite_sequence VALUES('Customer',13);
INSERT INTO sqlite_sequence VALUES('Staff',10);
INSERT INTO sqlite_sequence VALUES('MenuItem',10);
INSERT INTO sqlite_sequence VALUES('Reservation',10);
INSERT INTO sqlite_sequence VALUES('Orders',10);
PRAGMA writable_schema=OFF;
COMMIT;
```